PCI - AI

p53 TriControl™ · AI

p53 (TP53) immunohistochemistry · Sample report

Case ID	PCI-SAMPLE-0001	Specimen	Ovary, FFPE block	Marker	p53 (TP53)
Submitting site	Cedar Bend Hospital	Study diagnosis	High-grade serous carcinoma	Control lines	OE · WT · null on slide
Slides	H&E + p53 IHC	Tumor fraction	63 %	Calibration	Passed

p53 pattern	Strong (2+ or 3+) tumor nuclei	Stained (1+ or more)
Overexpression (abnormal pattern)	**94 %**	**99 %**

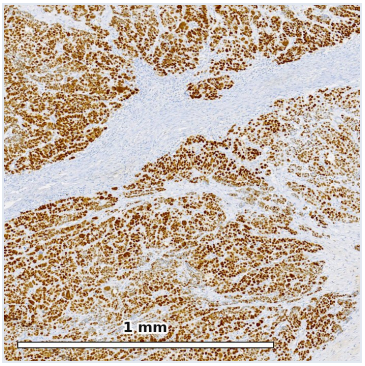
Reference ranges	Overexpression	Borderline	Null	Wild-type
	≥ 80 % of tumor nuclei strong (2+ or 3+) and ≥ 60 % stained	≥ 60 % stained but fewer than 80 % strong; pathologist review	≥ 80 % unstained (0)	otherwise

Results

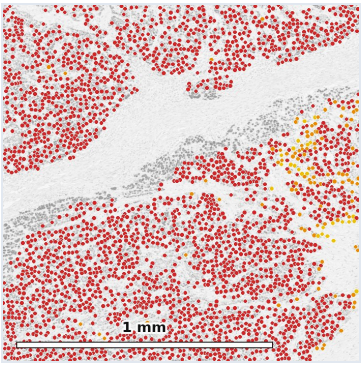
Grade	Stain	Share	Nuclei
0	Unstained	1 %	862
1+	Weak	5 %	3,180
2+	Moderate	10 %	6,646
3+	Strong	84 %	55,128
Tumor nuclei graded			**65,816**

Each tumor nucleus is graded against the three control lines stained on the same slide. The dot colours in picture (b) follow the grades in this table.

Tumor field



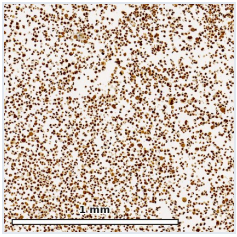
(a) p53 IHC, tumor field at about 10×.



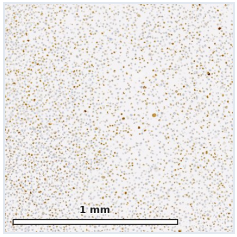
(b) The same field: one dot per tumor nucleus, coloured by grade.

On-slide controls

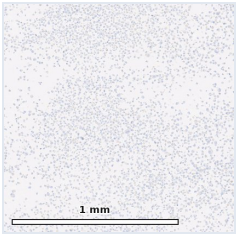
Three control cell lines of known p53 expression are carried on every slide and stained with the tissue. They set the grade scale for that slide, so a weak or strong run is read against its own controls.



OE · overexpression control



WT · wild-type control



null · null control

Assay description

p53 TriControl™ · AI reads a p53 immunohistochemistry slide against three characterized control cell lines carried on the same slide (reference 1): one that overexpresses p53, one with a wild-type level and one with no p53. The laboratory confirms the three control lines on the slide and the pathologist marks the tumor region; the software grades every tumor nucleus inside it from 0 to 3+ against that slide's own controls and scores it against established published p53 criteria (see References) and reports the pattern: wild-type, overexpression, null, or borderline for the pathologist's review. A pathologist reviews the slide, the controls and the AI's grades and signs the report. For Research Use Only. Not for use in diagnostic procedures.

References

1. Hussain I, Howard RS, Syed V, Allgäuer M, Gong H, Xing D, Andersen JD. A cell line–based immunohistochemical p53 expression pattern control panel. Int J Gynecol Pathol. 2019;38(5):449–458. doi:10.1097/PGP.0000000000000537
2. Köbel M, Piskorz AM, Lee S, Lui S, LePage C, Marass F, Rosenfeld N, Mes Masson AM, Brenton JD. Optimized p53 immunohistochemistry is an accurate predictor of TP53 mutation in ovarian carcinoma. J Pathol Clin Res. 2016;2(4):247–258. doi:10.1002/cjp2.53
3. Yemelyanova A, Vang R, Kshirsagar M, Lu D, Marks MA, Shih IeM, Kurman RJ. Immunohistochemical staining patterns of p53 can serve as a surrogate marker for TP53 mutations in ovarian carcinoma: an immunohistochemical and nucleotide sequencing analysis. Mod Pathol. 2011;24(9):1248–1253. doi:10.1038/modpathol.2011.85
4. Köbel M, Ronnett BM, Singh N, Soslow RA, Gilks CB, McCluggage WG. Interpretation of p53 immunohistochemistry in endometrial carcinomas: toward increased reproducibility. Int J Gynecol Pathol. 2019;38 Suppl 1:S123–S131. doi:10.1097/PGP.0000000000000488